English Translation: Japan's National Survival Strategy in the 2026 Complex Crisis: Brought by Resource Constraints and Autonomous AI

Introduction: A New Era Where Physical Constraints and Algorithmic Evolution Intersect

As of April 2026, the premises of the global economy and national security have been fundamentally overturned by irreversible paradigm shifts occurring simultaneously in two entirely different domains. The first shift is a severe physical resource supply disruption triggered by the de facto blockade of the Strait of Hormuz amid escalating conflict between Iran and the US/Israel.[1] The second shift is the collapse of premises in cybersecurity and system architecture, brought about by the limited release of Anthropic's state-of-the-art frontier AI model, "Claude Mythos Preview".[2]

The strategic proposition presented is that, in response to these two compulsory changes, Japan must seek survival not through incremental extensions of existing approaches, but through a fundamental redefinition of agricultural methods, system architectures, and diplomatic/technological leverage. This report rigorously verifies the provided market data, technical specifications, diplomatic trends, and AI benchmark results, comprehensively analyzing and integrating the validity of this strategic proposition and the overarching picture of concrete survival strategies Japan can adopt.

Part 1: Fragmentation of the Petrochemical Network and Redefinition of Food Security

1.1 The Dynamics of Naphtha Depletion and the Illusion of National Stockpiles

The destruction of petrochemical facilities in the Gulf region due to the Iran conflict and the drastic reduction in traffic through the Strait of Hormuz (a choke point responsible for about 25% of seaborne crude oil transport in normal times) pose a direct threat to Japan's industrial base.[1] The most crucial perspective in evaluating Japan's response to this crisis is the divergence between the macro stockpile of "crude oil" and the micro vulnerabilities in the "intermediate chemicals" supply chain.

As the data shows, Japan's crude oil stockpile itself appears rock-solid. Thanks to the strategic national stockpile system built on the lessons of the 1970s oil shocks, a total of

approximately 254 days' worth (or estimated between 230 to 228 days) of crude oil is secured, including about 143 days of national reserves.[3] Under the leadership of Prime Minister Sanae Takaichi, the release of about 30 days' worth of national reserves from late March, and an additional 20 days' worth from early May, has been decided, seemingly maintaining the surface-level energy supply.[4]

However, the situation for naphtha (crude gasoline), which can be called the "blood" of Japanese industry, is extremely dire. Naphtha is the basic raw material for all chemical products, including plastics, synthetic rubber, and medical components, yet Japan's dedicated domestic stockpile for naphtha is merely about 20 days' worth.[8] Even more fatal is the structural constraint that even if stockpiled crude oil is released to refineries, the extraction of fuel oils like gasoline and diesel to maintain the physical mobility of society is prioritized in the refining process.[8] With refinery utilization rates dropping to 67.7% of design capacity, increasing naphtha production is nearly physically impossible.[3] The combined buffer of naphtha-related product inventories, alternative procurement, and domestic refining is estimated at only about 4 months' worth. For Japan, which depends on the Middle East for 95% of its supply, this bottleneck is a ticking time bomb that could force the entire manufacturing sector to a halt.[9]

1.2 Vulnerabilities in the Fertilizer Supply Chain and Limitations of the Legal Framework

Alongside the naphtha shortage, the fertilizer supply network, which forms the foundation of food production, is also on the verge of collapse. Due to the halt in the supply of natural gas (LNG), the main raw material for nitrogen fertilizers, urea prices surged by 50% in just three weeks. Furthermore, the halt in sulfur supply is seriously impacting the production of phosphate fertilizers.

Amplifying this crisis is the institutional flaw in fertilizer stockpiling. Based on the Economic Security Promotion Act passed in 2022, fertilizers are designated as one of the specified critical materials (12 items).[11] However, unlike the physical "strategic national stockpile system" for oil, where the state owns and manages massive tanks, the fertilizer stockpile mechanism primarily relies on financial subsidies and compensation systems (such as risk mitigation measures through export credit agencies) for storage functions by private companies.[13] In a structure heavily dependent on overseas raw materials (especially from geopolitical risk areas like the Middle East and Russia), this private-dependent subsidy system becomes dysfunctional when a physical disruption of the supply chain occurs. Consequently, Japan's agricultural system is left extremely vulnerable to external shocks.

Resource Category	Stockpile Level (Estimated)	Form and Managing Entity of Stockpile System	Supply Vulnerability During Crisis

Crude Oil	Approx. 254 days	State-owned facilities and private tanks (physical stockpile)	Low (Stable in the short to medium term)
Naphtha	Approx. 20 days	Private inventory (dependent on crude refining process)	Extremely High (Risk of immediate depletion)
Fertilizer	Uncertain (Approx. several months)	Subsidies/Critical material designation for private storage	High (Structural depletion due to import interruption)
Intermediate Chemicals	Approx. 4 months	Retained inventory across the private supply chain	Medium (Chain shortages inevitable)

1.3 Limitations of Smart Farming and the Imperative Transition to Natural Farming

Under the dual constraints of fertilizer and fuel depletion, Japanese agriculture faces a structural problem where half-hearted compromises like "reduced pesticides and reduced chemical fertilizers" are impossible. Modern conventional agriculture is a system entirely dependent on petroleum products, from diesel fuel for tractors, vinyl mulch, pesticide spraying, and electric irrigation pumps to post-harvest packaging materials. If petroleum cannot be secured, the entire system cannot operate just by reducing fertilizers.

"Precision agriculture (smart farming)," frequently presented as an alternative, also cannot be a solution under the current crisis. Hardware such as drones, IoT soil sensors, autonomous tractors, and advanced irrigation networks rely heavily on naphtha-derived specialty plastics, polymers, and advanced electronic components for their manufacturing and maintenance.[15] In an environment where naphtha is depleted and the supply of resins and parts is cut off, smart farming relying on high-tech equipment is unsustainable.

Therefore, the only realistic and scientifically verified solution to the drastic reduction in fertilizer supply is a partial, forced transition to "Natural Farming (Zero Budget Natural

Farming: ZBNF / Community-Managed Natural Farming: CNF)," which yields harvests without using chemical fertilizers, pesticides, or petroleum-derived equipment. The proposed strategy of "first converting 30% of farmland to natural farming" functions as a rational risk hedge to drastically reduce reliance on external resources.

Data from large-scale empirical research involving hundreds of thousands of farmers in Andhra Pradesh, India, spearheaded by GIST Impact and the Global Alliance for the Future of Food, strongly supports the effectiveness of this transition. According to the research, the transition to natural farming did not lead to a decrease in yield but rather improved the yields of major crops like rice, maize, and millet by an average of 11%.[18] Furthermore, the elimination of purchasing costs for expensive chemical fertilizers and pesticides resulted in a dramatic 49% increase in farmers' net income.[18]

The secondary ecological effects are also immensely significant. Natural farming increases soil organic carbon by 5-18%, restores biodiversity, and substantially reduces health damage and medical costs due to pesticide exposure in rural areas.[18] While labor input increases by about 20%, this functions as a buffer to absorb local community employment as an alternative to high-cost mechanization.[18] Japan's temperate monsoon climate historically has high affinity with this type of farming, and by utilizing accumulated knowledge, it becomes possible to build a resilient hybrid food security model parallel to conventional agriculture (which concentrates limited fertilizers and fuels) on the remaining farmland.

Agricultural Model Indicator	**Conventional Agriculture (Chemical Intensive)**	**Natural Farming (ZBNF / CNF)**	**Strategic Evaluation**
External Resource Dependency	Extremely High (Fuel, imported fertilizer, pesticides)	Almost Zero (Local organic matter and microbes)	Establishing resilience during resource crises
Average Yield Change	Baseline	+11% Improvement (Major crops average)	Stabilization of food supply

Farmer Net Income	Baseline	+49% Increase (Due to reduced input costs)	Ensuring economic sustainability of agriculture
Soil and Environmental Impact	Carbon runoff, water pollution risks	Soil organic carbon +5-18%, zero pesticide use	Long-term restoration of ecological foundations

Part 2: The Mythos Shock and the Collapse of the Cybersecurity Paradigm

Coinciding with the physical supply chain collapse, an algorithmic singularity that fundamentally alters the design and philosophy of systems occurred in the digital realm. This is the emergence of Anthropic's latest AI model, "Claude Mythos Preview."

2.1 Project Glasswing and the Threat of Autonomous Exploits

"Project Glasswing," announced in April 2026, is a defense initiative where dozens of giant technology companies—including AWS, Apple, Google, Microsoft, Cisco, and CrowdStrike—have united to use the unreleased model Claude Mythos to explore and patch vulnerabilities in the world's critical infrastructure.[2] The reason this project was launched is the shocking fact that Mythos "has reached a level where it completely surpasses skilled human hackers in its ability to discover and exploit software vulnerabilities".[2]

Mythos's track record in vulnerability discovery destroyed the conventional wisdom of the security industry. Three points are particularly noteworthy:

First, in OpenBSD, which specializes in security and is lauded as "the world's most robust OS," it autonomously discovered a 27-year-old undetected vulnerability (integer overflow) in the TCP stack.[23] This bug could allow a system to be crashed remotely simply by connecting, slipping past years of audits by human experts. Furthermore, the computational cost required for this discovery was reportedly under $50.[27]

Second, the discovery of a 16-year-old vulnerability in FFmpeg, which is utilized in virtually all video processing systems. This code had been tested over 5 million times by automated fuzzing tools (a method of finding bugs by inputting massive amounts of random data) without any anomalies being detected.[2] By directly reasoning about the code's "semantics," Mythos identified logical flaws that blind testing could not find.

Third, the autonomous construction of a privilege escalation chain in the Linux kernel. Without any human instruction or guidance, Mythos autonomously chained multiple low-risk vulnerabilities to write a complex exploit overnight that hijacked root privileges (complete control of the system) from a general user.[23] In Firefox exploit benchmarks, while the previous generation Claude Opus 4.6 succeeded "2 times," Mythos recorded "181 times," achieving a non-linear performance improvement of about 90 times in a single generation.[27]

2.2 Reversal of the Economics of Defense and the End of the "Patch Paradigm"

The strategic implication indicated by these events is the fact that the "grace period from vulnerability discovery to exploitation" has been compressed to the absolute limit, from what used to be "months" to "minutes".[28]

Historically, cybersecurity has been an asymmetrical battle. Defenders had to protect all systems, while attackers only needed to find one loophole. However, until now, finding that "one loophole" required advanced expertise and vast amounts of time, which established the premise that vendors could provide "patches" in time.

If Mythos, or open-source lightweight models with equivalent capabilities (AISLE verification confirmed that even small-scale models could reproduce bug analyses similar to Mythos [27]) become widespread, this premise completely collapses. AI can discover thousands of zero-day vulnerabilities in a matter of weeks and autonomously generate exploits for all of them. In contrast, the traditional approach where human teams triage CVEs (Common Vulnerabilities and Exposures), verify them, and apply patches will be completely overwhelmed in both volume and speed.[31] In other words, the myth that "systems are safe until patched" has virtually vanished.

2.3 Forced Transition to In-house Operation (Self-contained) and the Pursuit of Simplicity

In a world where the cost of vulnerability discovery approaches zero and reactive measures through patches become impossible, enterprise and national system infrastructures are forced into a dramatic paradigm shift. This is the "radical simplification of systems" and a "return to insourcing (in-house operation) utilizing agent AI."

Complex system architectures—cloud infrastructure, integration of multiple SaaS, and multilayered development structures through system integrators (SIers) and consultants—themselves become massive attack surfaces. The more external libraries and intermediaries involved, the number of vulnerabilities lurking within the system increases exponentially, becoming targets for automated attack chains by AI.

As the proposed strategy outlines, now that AI agents of the Mythos class can execute everything from requirements definition, coding, testing, to dynamic security defense end-to-end, interposing intermediaries like SIers and cloud vendors is not just a waste of costs but an act that invites fatal security risks.

The optimal solution going forward is to build extremely small, self-contained systems with only the minimum necessary functions in-house, leveraging AI based on Open Source Software (OSS). The smaller and simpler the codebase, the higher the accuracy of constant monitoring and self-auditing by AI, physically reducing the room for unknown vulnerabilities to intervene. Eliminating complexity and maintaining a simple structure that you can "comprehend yourself and protect yourself" becomes the strongest security defense in the next generation. This shift to in-house operations is no longer a management option but a mandatory condition for survival.

Part 3: Japan's Survival Strategy in the New Era—Integration of Asymmetric Advantages

Facing the dual crises of physical resource depletion and the invalidation of cyber defense networks, Japan's path to survival lies entirely in the integrated utilization of three asymmetric advantages it possesses: "physical technologies that AI can never imitate," "historically built, person-dependent diplomatic channels," and "a hub function for logistics in Asia."

3.1 Physical Technology as a Black Box: MHI's J-Class Turbine and MGA1700

Even in an era where digital world codes are instantly analyzed and reconstructed by AI, advanced manufacturing technologies that rely on physical laws and the limits of materials science still function as indecipherable "black boxes." The foremost example is Mitsubishi Heavy Industries' (MHI) world-leading "J-Class Gas Turbine."

The J-Class turbine operates in an ultra-high temperature range of 1600°C to 1700°C, achieving an astonishing thermal efficiency of over 64% in Gas Turbine Combined Cycle (GTCC) power generation.[32] This efficiency is essential for extracting maximum power from limited natural gas fuel and dramatically reducing carbon dioxide emissions.[32]

The heart supporting this ultimate thermal efficiency is the turbine rotor blade made from the rhenium-free nickel-based single-crystal superalloy "MGA1700 (TMS-1700)," co-developed with the National Institute for Materials Science (NIMS).[34] The manufacturing technology of this alloy is not something simple that can be imitated just by having the blueprints. MGA1700 has extremely delicate properties; if even a few tens of ppm (parts per million) of sulfur (S) are mixed into the alloy, sulfur segregates at the interface of the protective oxide film (alumina scale) under high-temperature oxidizing environments, causing the film to peel and resulting in significant performance degradation.[35]

To prevent this fatal degradation, highly advanced metallurgical control is required during the melting process of manufacturing, such as using a calcium oxide (CaO) crucible or adding granular CaO to the molten alloy for desulfurization. This process must strictly reduce the sulfur concentration in the molten metal from an initial approx. 23-35 ppm down to a 2 ppm level capable of withstanding operation (capturing sulfur as CaS inclusions).[35]

Furthermore, the technology to cast turbine blades as a single crystal by controlling temperature gradients and speeds to the absolute limit during the solidification process is the fruit of decades of accumulated materials science and trial and error. Even if AI could understand the chemical formulas, it is impossible to instantly reproduce this macro physical casting equipment and micro impurity control know-how in the physical space.

3.2 Iran's Technological Limits and Japan's Bargaining Cards

The overwhelming performance of this J-Class turbine serves as the strongest diplomatic incentive toward current Iran. The MAPNA Group, which manages Iran's domestic power infrastructure, has successfully reverse-engineered Germany's Siemens older generation F-Class turbines (V94.3A, etc.) and indigenized them as "MGT-75" and "MGT-70".[39] This F-Class turbine has specs of a combustion temperature of about 1200°C to 1230°C and a combined cycle thermal efficiency of about 58% to 59%, which functions sufficiently for immediate recovery from destruction caused by conflict.[40]

However, for Iran to rebuild a highly advanced, internationally competitive petrochemical industry after sanctions are lifted, and to most efficiently convert its limited natural gas resources into foreign currency, the F-Class has its limits in terms of thermal efficiency and durability. If the Iranian leadership aims for the fastest and most efficient national reconstruction, introducing MHI's J-Class technology—currently the highest peak in the world—becomes the "optimal solution" they desperately desire.

Turbine Technology Comparison	**MAPNA Group MGT-75 (F-Class equivalent)**	**MHI J-Class Turbine**	**Strategic Differences & Advantages**
Base Technology	Siemens V94.3A (Reverse Engineering) [39]	In-house Development / Culmination of National Projects [43]	Japan's complete intellectual property and technological independence
Turbine Inlet Temp (TIT)	Approx. 1230°C [41]	1600°C-1700°C [32]	Achieving overwhelming combustion efficiency
Combined Cycle Efficiency	Approx. 58%-59% [40]	Over 64% [32]	Massive reduction in fuel consumption and CO2 emissions

Blade Metallurgy Tech	Conventional single crystal / polycrystal	MGA1700 single-crystal superalloy (Extreme desulfurization control) [35]	Black-box manufacturing network impossible for AI to imitate

3.3 Irreplaceable Diplomatic Channels and Exceptional Securing of the Strait of Hormuz

Japan's second advantage is an irreplaceable, historically thick, and person-dependent diplomatic pipeline to convert this physical technology into a diplomatic victory. The current Iranian Foreign Minister, Abbas Araghchi, served as the Iranian Ambassador to Japan from 2008 to 2011 and has experience as a visiting professor for the Executive Management Program (EMP) at the University of Tokyo, making him an exceptionally pro-Japanese diplomat.[44] His doctoral thesis, which themed the harmonization of Western democratic theory and Islamic principles, also indicates his high realistic and practical coordination capabilities within the Iranian regime.[44]

This pipeline is demonstrating dramatic effects in the emergency situation under conflict. While the Islamic Revolutionary Guard Corps (IRGC) targets vessels related to the US and Israel, effectively closing the strait, Foreign Minister Araghchi issued an exceptional signal to Japan. In an interview, he clearly stated, "The strait is closed only to ships belonging to our enemies. For other countries, ships can pass through, and we are talking to them (Japan) to find a safe way to pass".[49]

Prime Minister Takaichi capitalized on this opportunity and immediately held a summit-level telephone conference with Iranian President Pezeshkian.[52] The cards Japan must play here are clear. By presenting incentives such as providing the highest-performance gas turbine technology (J-Class) and post-war reconstruction support, Japan must secure guarantees for the safe passage rights (securing sea lanes) of Japan-related vessels in the Strait of Hormuz in return. This is a unique deal that only Japan can execute among Western allies.

3.4 Evolution into the Coordinator (Hub Function) of the Pan-Asian Network

Even if exceptional transit rights through the Strait of Hormuz are secured, that alone cannot prevent the collapse of supply chains across Asia surrounding Japan. As China and South Korea move to restrict exports prioritizing their own countries, Japan must function as a "Pan-Pacific hub" to strategically redistribute the secured energy.

While the dollar-denominated international market mechanism becomes dysfunctional, Japan has already embarked on building a physical barter energy swap network. Examples include naphtha and LPG swap transactions with India, LNG and LPG exchanges with Indonesia, and support for the Philippines and Vietnam through emergency fuel supplies (agile supply networks including mobile dispensers).[54]

These transactions are made possible by the vast domestic network of stockpile bases and sophisticated LNG/LPG receiving and exporting infrastructures that Japan has built through massive investments over many years. By functioning as a "router (distributor)" of physical energy during a crisis despite being a net energy importer, Japan can prevent a cascading economic collapse across Asia, and consequently maintain its own manufacturing supply chains (such as parts supply networks).

Conclusion: Towards the Establishment of Autonomous Resilience

The conclusion suggested by the 2026 complex crisis is clear. The "peacetime premises" of existing global supply chains and IT security reliant on third parties have already perished. For Japan to survive this new era, a three-tiered paradigm shift is essential.

1. **Decarbonization of Food and Resource Infrastructure:** The depletion of naphtha and fertilizers is inevitable. To counter this, 30% of agriculture must be forcibly transitioned to "Natural Farming (ZBNF/CNF)," drastically reducing dependence on imported chemical fertilizers and fossil fuels. This is not merely an environmental policy but a cold, hard decision for national security.
2. **Miniaturization and In-house Operation of IT Infrastructure:** With the advent of autonomous AI represented by Claude Mythos, "complex systems" have become an indefensible liability. It is urgently necessary to eliminate SIers and cloud intermediaries and transition to an architecture where minimum necessary, autonomous, self-contained systems using open source are operated and defended in-house.
3. **Strategic Diplomatic Deployment Leveraging Asymmetric Technology:** The ultimate collateral for Japan's survival is "deep physical technologies that AI cannot copy and other nations cannot substitute," such as MHI's J-Class turbine and MGA1700 superalloy. By connecting this with the unique diplomatic channel of the Araghchi route, Japan must establish its position as a coordinator controlling the bloodstream of energy across Asia.

Whether these strategies can be executed in an integrated and agile manner will determine the survival of Japan and all of Asia in the new era.

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